



Anomalous Aortic Origin of a Coronary Artery: Preoperative Diagnosis and Surgical Planning

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Abstract

Background: Anomalous aortic origin of a coronary artery (AAOCA), the anomalous coronary artery arises from an inappropriate coronary sinus and travels between the aorta and pulmonary artery. Proper surgical management depends upon correct diagnosis and accurate characterization of the origin and course of the coronary artery. Transthoracic echocardiography (TTE) has been the mainstay for diagnosis, but magnetic resonance imaging (MRI) and computed tomographic angiography (CTA) have been increasingly utilized. In this study, we report the largest series of surgically repaired AAOCA and accuracy of preoperative diagnostic studies. **Methods:** A review of 53 consecutive patients (mean age 13.9 years, range 4-65 years) undergoing repair of an AAOCA from 1995 to 2009 was performed. In all, 40 patients were identified with an anomalous right coronary artery (ARCA) from the left sinus of Valsalva, 13 patients had an anomalous left coronary artery (ALCA) arising from the opposite sinus. Symptoms of angina or syncope were present in 58% and 46% of cases with ARCA and ALCA, respectively. Results of preoperative diagnostic testing were compared to actual surgical findings to determine the accuracy of the tests. **Results:** Lack of an intramural course was observed intraoperatively in 7 cases (5 ARCA and 2 ALCA). Preoperative TTE accurately predicted whether the AAOCA was intramural or extramural in 49 (92.5%) of 53 cases. Magnetic resonance imaging was predictive in 5 (83.3%) of 6 patients and CTA in 11 (64.7%) of 17. Survival was 100%. Complications occurred in 4 (7.5%) of 53 patients (mean follow-up 29 months). Patency was confirmed in 97.7% with TTE, and 23 (95.8%) of 24 patients had a negative postoperative functional study. **Conclusions:** Transthoracic echocardiography was found to be very accurate at defining the presence or absence of an intramural course in AAOCA. Both MRI and CTA can provide additional information but may not be as accurate as TTE.

Keywords

artery/arteries, congenital heart surgery, congenital heart disease, coronary artery anomaly, coronary artery imaging (all modalities), coronary artery surgery, echocardiography

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Introduction

Anomalous aortic origin of a coronary artery (AAOCA) from the incorrect sinus of Valsalva has an estimated prevalence of 0.17% to 1.2% based on numerous autopsy, echocardiography, and coronary arteriography studies.^{1,2,3} In its most common form, the anomalous artery takes an interarterial course between the aorta and main pulmonary artery and travels within the wall of the aorta. The origin of the anomalous artery frequently has a slit-like orifice with an acute angle of exit from the aorta. In some instances, the anomalous artery may arise from a common orifice with the opposite coronary artery or may arise as a single artery. These abnormalities of coronary artery origin and course have been implicated in exercise-induced sudden death in children and young adults.^{4,5} The etiology of sudden

death may involve cardiac fibrosis and ventricular arrhythmias secondary to repeated subclinical episodes of coronary ischemia.

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Abbreviations and Acronyms

AAOCA	anomalous aortic origin of a coronary artery
ALCA	anomalous left coronary artery
ALMCA	anomalous left main coronary artery
ARCA	anomalous right coronary artery
CABG	coronary artery bypass grafting
CHF	congestive heart failure
CTA	computed tomographic angiography
MRI	magnetic resonance imaging
RIMA	right internal mammary artery
TTE	transthoracic echocardiography

Postulated mechanisms of ischemia include ostial obstruction due to a slit-like coronary orifice, compression of the anomalous coronary between the great vessels, anomalous coronary vasospasm, dynamic flattening or kinking of the anomalous coronary with aortic distention, or obstruction of the intramural segment with aortic expansion, often at the level of the commissure.^{5,6}

Anomalous aortic origin of a coronary artery with anomalous left main coronary artery (ALMCA) from the right sinus of Valsalva has increased risk of sudden death and adverse cardiac events, significantly higher than for AAOCA with anomalous right coronary artery (ARCA) from the left sinus.^{1,3,4} Premonitory symptoms of chest pain and syncope appear in only 37% of sudden death victims. Conversely, 63% of AAOCA-attributed deaths occur in previously asymptomatic individuals. Consequently, it has been our institutional practice to recommend repair for all symptomatic AAOCA and asymptomatic anomalous left coronary artery (ALCA) at the time of diagnosis. Management of children with asymptomatic ARCA remains more controversial.

Further complicating the management of AAOCA is the absence of a reliable risk stratification based on preoperative testing. Analyses of anomalous coronary morphology, electrocardiography, and provocative functional tests have failed to correlate with the risk of sudden death.^{6,7} By virtue of the guidelines set by the 36th Bethesda Conference, any patient with an AAOCA should be excluded from all competitive sports.^{8,9} For patients with ARCA who wish to participate in strenuous athletics, we recommend surgical intervention. It is not entirely clear that a successful surgical correction completely alleviates the risk of cardiac events and the long-term outcome of these patients is not known at this time.¹² Therefore, the surgical repair must be very low risk to justify such a recommendation. Proper selection of an appropriate surgical procedure depends on accurate diagnostic identification of the anomaly. In this study, we report the largest single series of surgically repaired AAOCA, highlighting the accuracy of preoperative imaging.

Methods

A review of the Duke University Congenital Cardiovascular Surgery database from April 1995 to January 2010 identified all

patients with AAOCA who underwent surgical repair. The Duke University Medical Center Institutional Review Board approved this study. A total of 53 patients were identified with AAOCA, in which the coronary artery ostium originated in the opposite sinus and followed an interarterial course between the aortic root and main pulmonary artery. Patients without an interarterial course were excluded. The medical records of these patients were reviewed, including patient demographics, presenting symptoms, preoperative and postoperative imaging, operative reports, complications, and outcomes. Preoperative imaging results were compared to intraoperative surgical findings to determine the accuracy of the tests. Patency of the newly repaired coronary artery was determined by transesophageal echocardiogram. Functional studies after repair included stress test (treadmill), cardiac magnetic resonance imaging (MRI) with adenosine, and dobutamine stress echocardiography. Confirmation of exercise restriction removal was obtained at last follow-up visit with cardiologist.

Surgical Technique

All operations were performed through a median sternotomy with distal aortic and right atrial cannulation. Standard cardiac surgical techniques utilizing mild hypothermic cardiopulmonary bypass (32°C-34°C) and antegrade cardioplegia only were utilized. In each case, a transverse aortotomy was performed following cardiac arrest. The interarterial morphology dictated the surgical approach. Patients with an extramural course of the anomalous vessel underwent coronary translocation and reimplantation into the appropriate sinus. In patients with an intramural component to the anomalous coronary artery, patients underwent a simple unroofing procedure if the coronary artery originated at, or distal to the intercoronary commissure.^{10,11} If the intramural segment passed proximal to the commissure, within the sinus, the patient underwent either neostial creation or coronary unroofing, with detachment and resuspension of the commissure. Only 1 patient underwent coronary artery bypass grafting (CABG) with an in situ right internal mammary artery (RIMA) for an ARCA. In this patient, the ARCA arose from the left coronary and traveled in an interarterial course without an intramural segment. The operating surgeon felt that the proximal right coronary artery was too small for translocation and reimplantation and chose to perform a RIMA to the RCA bypass graft. Another patient with repair of an ALCA underwent typical unroofing procedure but also had a proximal RCA atherosclerotic lesion and underwent CABG with a RIMA to RCA.

Statistical Analysis

All data were obtained for this study by medical record review. Continuous variables obtained in this study are expressed as mean $\pm$ standard error of the mean or as means with ranges.

Results

Fifty-three patients underwent surgical repair of AAOCA. Patient demographics are presented in Table 1. The mean age

Table 1. Patient Characteristics

Characteristic	Mean (Range)
Age (years)	13.9 ± 12.0 (4-65 years)
Sex	
Male	39/53 (73.5%)
Female	14/53 (26.4%)
Symptoms at presentation	
ALCA (angina or syncope)	6/13 (46.2%)
ARCA (angina or syncope)	23/40 (57.5%)
Mean follow-up	29.03 months

Abbreviations: ALCA, anomalous left coronary artery; ARCA, anomalous right coronary artery.

Table 2. AAOCA Morphology

Morphology	N (%)
ALCA	13/53 (24.5) ^a
Intramural	11 (84.6)
Extramural	2 (15.3)
ARCA	40/53 (75.4) ^a
Intramural	35 (87.5)
Extramural	5 (12.5)

Abbreviations: AAOCA, anomalous aortic origin of a coronary artery; ALCA, anomalous left coronary artery; ARCA, anomalous right coronary artery.

^a Percentage of total.

Table 3. Definition of Intramural Course: Accuracy of Preoperative Imaging

Imaging Modality	N (%) ^a
TTE	49/53 (92.5)
CTA	11/17 (64.7)
MRI	5/6 (83.3)

Abbreviation: CTA, computed tomographic angiography; MRI, magnetic resonance imaging; TTE, transthoracic echocardiography.

^a Percentage predictive of interarterial morphology based on intraoperative visualization and preoperative imaging modality obtained.

of patients undergoing surgical correction was 13.9 years (range 4-65 years). Forty patients were identified with an ARCA arising from the left sinus and 13 patients had an ALCA arising from the right sinus. Symptoms were present in 46% of patients with ALCA and in 57% of those with ARCA. Presenting symptoms included syncope and angina, with 1 patient with ARCA experiencing an episode of sudden death.

All of the patients in this series had AAOCA with an interarterial course. The course of the anomalous coronary was further characterized by the presence or absence of an intramural course (Table 2). Preoperative imaging was undertaken for all patients with AAOCA and is shown in Table 3. All patients were evaluated preoperatively with transthoracic echocardiography (TTE). Computed tomographic angiography (CTA) was performed in 17 patients and MRI in 6 patients. In the latter part of the series, confirmatory testing with either MRI or CTA was

Table 4. Operative Procedures^a

Operative Procedure	N (%)
Modified unroofing	43/53 (81.1)
Coronary translocation	6/53 (11.3)
Neoostial creation	3/53 (5.7)
CABG	1/53 (1.9)

Abbreviation: CABG, coronary artery bypass grafting.

^a Percentage predictive of interarterial morphology based on intraoperative visualization and preoperative imaging modality obtained.

Table 5. Surgical Outcomes^a

Outcome	N (%)
Survival	53/53 (100)
Complication rate (AI, PTX, pleural effusion, bleeding, retained foreign body)	5/53 (9.4)
Postoperative TTE patency	43/44 (97.7)
Postoperative symptomatic relief ^a	35/43 (81.4)
Postoperative exercise restriction removal	36/40 (90)

Abbreviations: AI, aortic insufficiency; PT, pneumothorax; TTE, transthoracic echocardiography.

^a There was no mortality in these 53 patients with 5 complications. Of the 8 patients who had some symptoms after surgery, only 4 had symptoms potentially consistent with cardiac ischemia and had negative functional testing for ischemia.

routinely performed. Early in the series, these tests were reserved for cases in which transthoracic imaging was not diagnostic. In all patients, the presence of an interarterial course was accurately predicted by the preoperative imaging. Transthoracic echocardiography accurately predicted the presence or absence of the intramural course 92.5% of the time. Both MRI and CTA scans were not as effective at accurately characterizing the course (83.3% and 64.7%, respectively).

Operative interventions and results are outlined in Table 4. Operations included 43 unroofing procedures, 3 neoostial creations, 6 coronary translocations, and 1 CABG. Associated procedures were performed in 4 patients, including an Ebstein anomaly repair, atrial septal defect closure, pulmonary valvotomy, and CABG. Lack of an intramural course was observed intraoperatively in 7 cases (5 ARCA and 2 ALCA). There was no operative mortality in this series and the mean follow-up was 29 months. Perioperative complications occurred in 7.5% (4 of 53) of patients, including pneumothorax, pleural effusion, bleeding requiring reoperation, aortic insufficiency requiring intervention, and a retained foreign body. Of the 53 patients, 24 underwent functional study assessment to evaluate potential ischemia after repair (Table 5). These included exercise treadmill stress test, adenosine MRI stress test, and dobutamine stress echocardiography. Some of these were performed for symptoms and some as a routine follow-up; 95.8% (23 of 24) had a negative study. The 1 patient who had a positive test had a widely patent right coronary artery on subsequent coronary angiography. All patients, including patients with asymptomatic ARCA, were placed on exercise limitation at the time of

their diagnosis. Exercise restriction was withdrawn after surgery by the referring cardiologist in 37 of 40 patients in whom the cardiologist commented. In the 3 patients who remained on exercise limitation, 2 had documented postoperative chest pain as the reason for continued restriction, despite the fact that both had negative stress tests. The third patient had no symptoms and the reason for exercise limitation was not clear.

Of the 43 patients, 8 had symptoms of potential cardiac dysfunction postoperatively (6 with chest pain, 1 with mild congestive heart failure [CHF], and 1 with syncope). All patients with postoperative symptoms had the diagnosis of ARCA. Of the 8 patients 7 had an unroofing procedure, while 1 had coronary translocation. Importantly, 7 of the 8 patients who had symptoms postoperatively were symptomatic preoperatively. Of the 6 patients with chest pain, only 4 were suspicious for potential cardiac etiology. Of these 4 patients, 3 had a stress test postoperatively, of which all were negative studies. The fourth was only 1 month from surgery at the time of this report and had not undergone a stress test. The patient with postoperative syncope had a negative MRI adenosine stress test. In the 1 patient with mild CHF, the exercise treadmill test was positive and subsequent cardiac catheterization revealed a widely patent right coronary artery. No patient in this series had myocardial perfusion scanning.

Discussion

Anomalous aortic origin of a coronary artery is considered a relatively rare cardiac defect. Anomalous left main coronary artery and ARCA have an estimated incidence of 0.03% to 0.05% and 0.1%, respectively.^{1,2,3,15} Both coronary lesions have some risk of cardiac complications including angina, syncope arrhythmia, and sudden death with ALMCA having a significantly greater risk than ARCA.^{1,13,14} There exists a general consensus that all symptomatic anomalies, as well as asymptomatic ALMCA, should undergo surgical correction. However, debate exists regarding the management of the asymptomatic ARCA.¹⁶ There are recommendations from the 36th Bethesda Conference for the management of any congenital coronary artery anomaly stating that "detection should result in exclusion from all participation in competitive sports." This is an unacceptable situation for many patients and parents who want their children to be able to compete in competitive sports. Faced with this dilemma, many groups offer surgery for asymptomatic ARCA even though the risk of sudden death is relatively low, estimated at 0.2% at 20 years.¹⁶ So if operation is to be recommended in this situation, the operation should be as effective and low risk as possible. In our series, there was no mortality and minimal morbidity with the surgical repair. However, in our series, 8 of the 53 patients experienced symptoms postoperatively that may have been cardiac in nature. However, upon evaluation, only 4 of the 8 patients had symptoms that were possibly cardiac in origin. Of the 8 patients, 6 had functional testing and all were negative. In total, 24 of the 53 patients had postoperative functional testing for ischemia. Only 1 of these was positive in a patient with heart failure symptoms.

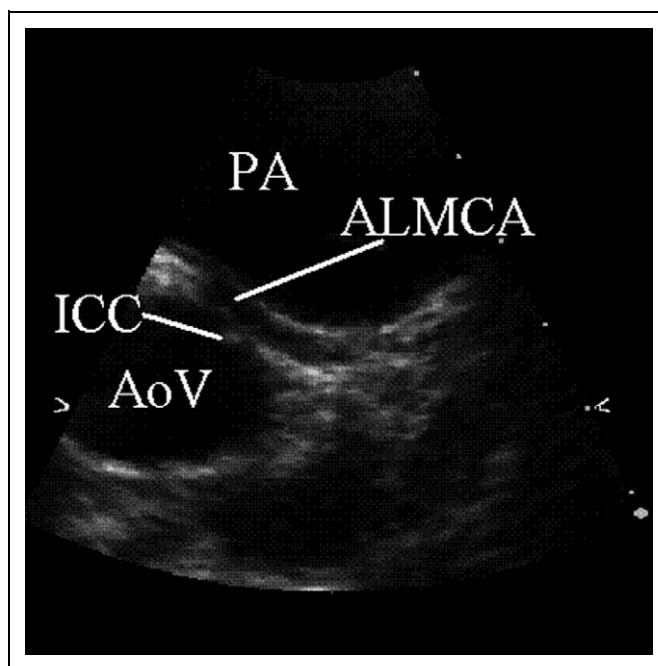


Figure 1. A transthoracic image of a patient with an anomalous aortic origin of the left main coronary artery (ALMCA). The intramural course is between the aorta (AoV) and pulmonary artery (PA) and the caliber of the intramural artery is less at the level of the intercoronary commissure (ICC).

This patient subsequently had a widely patent coronary on angiography. Recognizing the limitations of these functional tests, based on these data, we have routinely released patients for unrestricted exercise after successful surgical repair.

This report describes the largest reported institutional series of surgically repaired AAOCA to date, with a special emphasis on defining the best diagnostic imaging study. Preoperative knowledge of the interarterial morphology remains imperative to operative planning and positive surgical outcomes in patients undergoing AAOCA repair. Preoperative imaging modalities (TTE, CTA, or MRI) were assessed for accuracy in predicting interarterial morphology (ie, the presence or absence of an intramural segment). Radiology reports were compared to operative findings in making this assessment. Transthoracic echocardiography (Figure 1) was performed as the initial diagnostic examination in all 53 patients undergoing surgical repair and accurately defined the presence or absence of an intramural course in 92.5%. Improvements in ultrasound technology and the combination of TTE with color Doppler flow in prospective studies have shown identification of both the ALCA and ARCA.^{17,18} In some cases, 2-dimensional echocardiography can be misleading especially in patients who have AAOCA and an intramural course because the anomalous coronary can appear to arise from the correct sinus. With the addition of color Doppler flow, the direction of the flow can be evaluated in the intramural segment and help with the differentiation of the sinus of origin. However, echocardiography is still dependent on the patient's anatomy and acoustic window and limited ability to detect the orifice of the coronary artery.

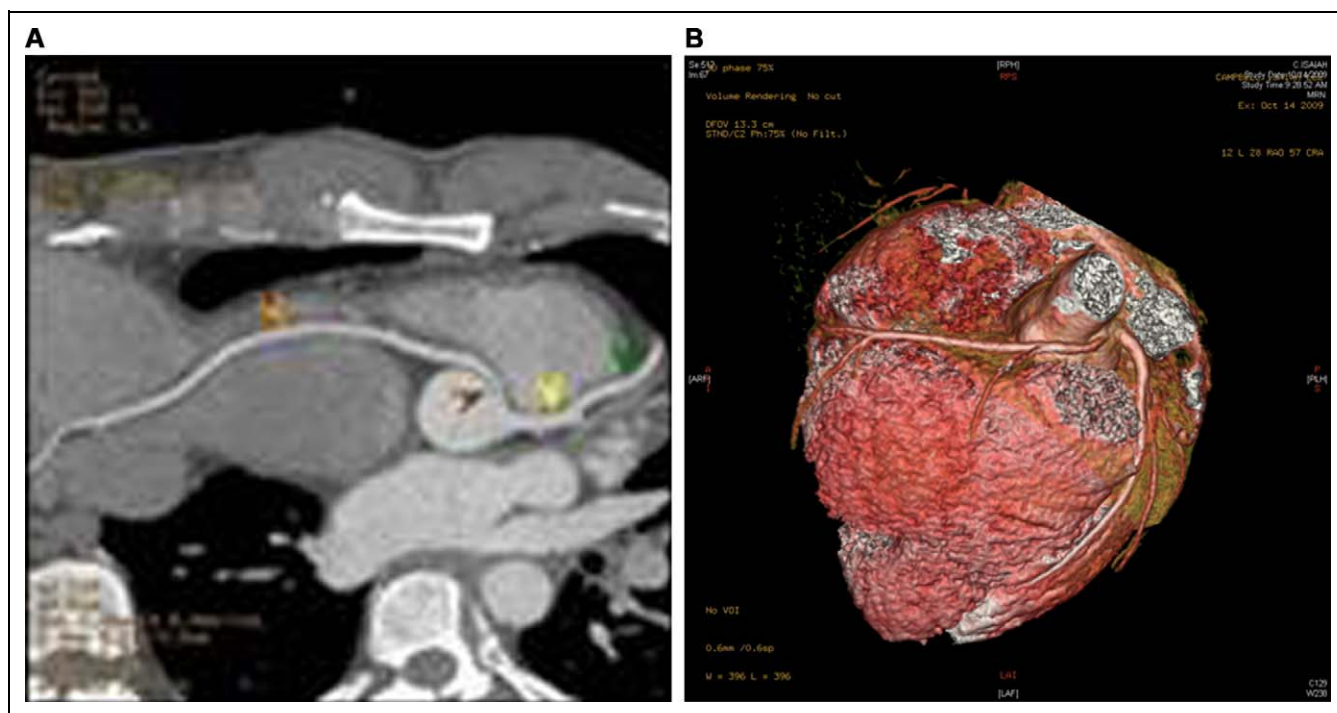


Figure 2. A and B, Computed tomography scan reconstruction of a patient with AAOCA ARCA. This patient was thought to have an intramural course of the anomalous artery but at the time of surgery, the artery was extramural and the patient underwent a coronary artery translocation into the appropriate sinus. AAOCA indicates anomalous aortic origin of a coronary artery; ARCA, anomalous right coronary artery.

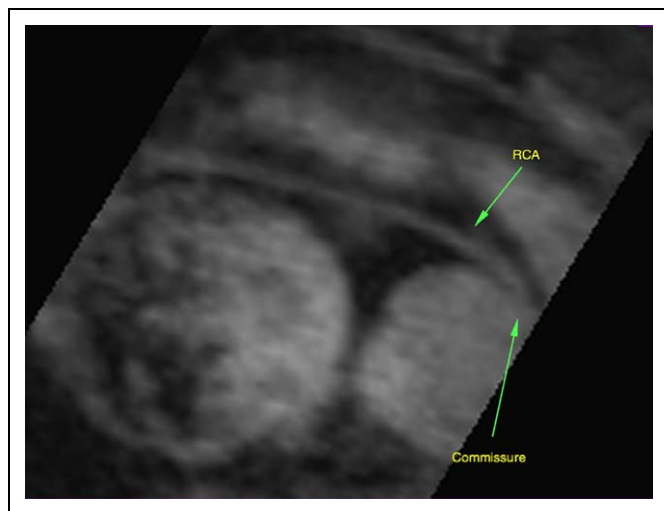


Figure 3. Magnetic resonance image of a 15-year-old female competitive soccer player who had chest pain and near syncope with exertion. She had a very short intramural segment of the anomalous right coronary artery (ARCA) that arose just at the level of the intercoronary commissure (ICC). She underwent a coronary artery translocation.

Computed tomographic angiography correctly predicted the presence or absence of an intramural course in only 64% of the cases reviewed in our series. Computed tomography angiography is a noninvasive, nonoperator-dependent technique with radiation comparable to standard angiography that provides high spatial and temporal relationships of the heart and coronary vasculature (Figure 2A, B). CTA delineates the

AAOCA and its relationship to the great vessels and other cardiac structures similar to that used with coronary angiography. CTA also has the ability to define the degree of compression of the intramural ectopic vessel.¹⁹ However, no study has reported the accuracy of CTA in defining the intra- or extramural course of the AAOCA.

In this series, MRI accurately predicted the intramural course in 83% of the cases in which it was used (Figure 3). Magnetic resonance imaging resolution accurately delineates the course of the AAOCA. Magnetic resonance imaging also offers the ability to evaluate the functional significance of anatomical findings by combining stress testing. Magnetic resonance imaging has previously been compared to conventional angiography in a study by Bunce.²⁰ Magnetic resonance imaging accurately described the interarterial course in 6 of 6 patients with ARCA, while conventional angiography identified only 3 of 6. Three-dimensional magnetic resonance angiography accurately defined the course of the ALMCA while the course of the ALMCA was uncertain in 6 of 9 by conventional angiography.²¹

Limitations

This study is a retrospective observational study. The TTE, CTA, and MRI examinations were not prospectively evaluated. In our study, CTA and MRI were used as either confirmatory tests or in patients in whom the course of the coronary artery was more difficult to define by TTE. Thus, the true accuracy of these tests may be underestimated. Therefore, statistical analysis of these differences is not reported. Despite this, we recommend

TTE as the initial diagnostic and screening modality. Magnetic resonance imaging is our first-choice test to confirm or further define the morphology of the anomalous coronary.

Conclusion

In our experience, surgical correction of AAOCA can be performed safely and effectively. Survival is excellent, however, there can be complications and successful repair depends upon accurate diagnosis and definition of course of the interarterial coronary artery. Transthoracic echocardiography is very accurate at defining the presence or absence of the intramural course of the coronary artery. Both CTA and MRI may be somewhat less accurate but may provide additional useful information. Controversy still exists regarding the management of the asymptomatic ARCA. Because of the existing recommendations for exercise limitations, surgeons and cardiologists may recommend surgery, even though there may be very low risk of adverse outcome without surgery.

Many of the unanswered questions regarding the surgical and nonsurgical history of AAOCA may be addressed through the multi-institutional registry established by the Congenital Heart Surgeons Society and the Anomalous Coronary Artery Working Group. With further understanding of the natural history and surgical effectiveness, rational guidelines regarding the risk of AAOCA and exercise restrictions may be adopted.

Declaration of Conflicting Interests

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